

MINGYU PARK

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RESEARCH INTERESTS

Challenging yet promising research aimed at:

- [1] Discovering new chemical tools to probe and control biological systems.
- [2] Unveiling oxidation-driven biological mechanisms underlying human disease.
- [3] Leveraging chemical approaches to address unmet clinical needs.

CURRENT POSITION

Postdoctoral Fellow

Mar. 2025 – Present

funded by NRF of Korea, Basic Science Research Program

Ulsan National Institute of Science and Technology (UNIST), South Korea

Projects: scalable photooxidation, LLPS, ferroptosis

EDUCATION

M.S./Ph.D. in Chemistry

Sep. 2018 – Feb. 2025

Ulsan National Institute of Science and Technology (UNIST), South Korea

Supervisor: Prof. Tae-Hyuk Kwon

Thesis: "Molecular and biological mechanism studies of protein photooxidation to control cell fate and their therapeutic applications"

B.S. in Chemistry

Mar. 2014 – Aug. 2018

Ulsan National Institute of Science and Technology (UNIST), South Korea

PUBLICATIONS

First-author articles

† Equal contribution.

- [1] "Rational Design of Biocompatible Ir(III) Photosensitizer to Overcome Drug-Resistant Cancer via Oxidative Autophagy Inhibition"
Mingyu Park†, Jung Seung Nam†, Taehyun Kim†, Gwangsu Yoon, Seoyoon Kim, Chaiheon Lee, Chae Gyu Lee, Sungjin Park, Kochan S. Bejoymohandas, Jihyeon Yang, Yoon Hee Kwon, Yoo Jin Lee, Jeong Kon Seo, Duyoung Min*, Taiho Park*, and Tae-Hyuk Kwon*
Adv. Sci. **2025**, **12**, 2407236
Conceptualization, preparing manuscript, contributing all aspects of the research.
- [2] "Hidden route of protein damage through oxygen-confined photooxidation"
Seoyoon Kim†, Eojin Kim†, **Mingyu Park†**, Seong Ho Kim†, Byung-Gyu Kim, Victor W. Sadongo, W.C. Bhashini Wijesinghe, Yu-Gon Eom, Gwangsu Yoon, Chaiheon Lee, Hannah Jeong, Chae Un Kim, Kyungjae Myung, Jeong-Mo Choi, Seung Kyu Min*, Tae-Hyuk Kwon*, and Duyoung Min*
Nat Commun **2024**, **15**, 10873
Preparing manuscript, establishment and experimental verification of the hypothesis

- [3] "Oxidative photocatalysis on membranes triggers non-canonical pyroptosis"
Chaiheon Lee†, **Mingyu Park†**, W. C. Bhashini Wijesinghe†, Seungjin Na, Chae Gyu Lee, Eunhye Hwang, Gwangsu Yoon, Jeong Kyeong Lee, Deok-Ho Roh, Yoon Hee Kwon, Jihyeon Yang, Sebastian A. Hughes, James E. Vince, Jeong Kon Seo*, Duyoung Min*, and Tae-Hyuk Kwon*
Nat Commun 2024, 15, 4025
Initiation, preparing manuscript, leading proteomic analysis, analyzing chemical/biological mechanisms

Co-author articles

- [4] "Zero-length photo-crosslinking by organoiridium catalyst for intracellular interactome mapping"
Jung Seung Nam, Myeong-Gyun Kang, Chang-Mo Yoo, Chae Gyu Lee, **Mingyu Park**, Jeong Kon Seo*, Hyun-Woo Rhee*, Tae-Hyuk Kwon*
under revision
Spectroscopic analysis of protein–catalyst interactions, interpreting labeled proteome
- [5] "Analysing the mechanism of mitochondrial oxidation-induced cell death using a multifunctional iridium(III) photosensitiser"
Chaiheon Lee, Jung Seung Nam, Chae Gyu Lee, **Mingyu Park**, Chang-Mo Yoo, Hyun-Woo Rhee, Jeong Kon Seo*, & Tae-Hyuk Kwon*
Nat Commun 2024, 12, 26.
Transient spectroscopic analysis of a probe in a viscous environment

GRANTS

- [1] **(PI)** Basic Science Research Program – National Research Foundation of Korea
"Scalable photooxidation for next-generation photodynamic therapy"
Sep. 2025 – Aug. 2028
- [2] (Participating) Basic Research in Science and Engineering – National Research Foundation of Korea
"Development of organic photocatalyst based on water splitting for high resolution protein-interactome mapping"
Mar. 2025 – Feb. 2028
- [3] (Participating; *Selected as 2025 Outstanding results in healthcare R&D*)
National R&D Program for Cancer Control – National Cancer Center
"Endomembrane-target oxidation of pancreatic cancer for immune activation"
Apr. 2022 – Dec. 2024
- [4] (Participating) Tech Incubator Program for Startup – Korea Technology and Information Promotion Agency for SME
"Development of a Photodynamic Therapy (PDT) Platform Using Next-Generation. Photosensitizers"
Sep. 2021 – Aug. 2023

SKILLS

Independent Research

Writing grant proposals, designing projects, adapting alternative plans, preparing manuscripts, mentoring junior researchers, presentation, and engaging in interdisciplinary/collaborative works.

Molecular Design and Synthesis

Design and synthesis of photoresponsive organic/organometallic compounds with diverse energy levels for therapeutic applications. DFT calculations using Gaussian.

Characterization

Characterization using NMR, UV–Vis, steady-state/transient photoluminescence, IR spectroscopy, HPLC, MALDI-TOF, and LC-MS. Electrochemical analysis by cyclic voltammetry and chronoamperometry.

Biochemical Techniques

2D/3D cell culture, cytotoxicity assays (MTT, LDH, Live/Dead assay), ROS detection assays, SDS and SDS-free gel electrophoresis of proteins, immunoblot, gene transfection, immunofluorescence staining, confocal microscopy of fixed/live cells or tissue samples.

Proteomic sampling and analysis

In-gel and on-filter proteomic sampling for LC-MSMS. Analysis and visualization of the identified proteome (FragPipe, Cytoscape, Gephi) and protein structures (PyMOL).

AWARDS

- Best Poster Award - 21st Congress of the European Society for Photobiology (Aug. 2025)
- 2025 Graduate Student Excellent Research Award (Feb. 2025)
- Poster of Excellence - 2024 Korean Chemical Society Fall Meeting (Oct. 2024)

PATENT

- Tae-Hyuk Kwon, Chaiheon Lee, and **Mingyu Park**
"WATER SPLITTING BASED PHOTOSENSITIZER AND COMPOSITION. FOR. PHOTODYNAMIC THERAPY"
Korea Patent - Application No. 10-2023-0051598)

TALKS

- **(Invited)** International Photodynamic Association: Rising Stars in PDT Research (May 2025)
- The Polymer Society of Korea 2025 Spring Meeting (Apr. 2025)
- 129th General Meeting of the Korean Chemical Society (Apr. 2022)

TEACHING

Academic Service

Lab-based Mentorship

Sep. 2024 – Present

Led weekly lab meetings and journal club; provided guidance on experimental design and data analysis for 6 graduate students, 3 of whom received awards at academic conferences.

Teaching Assistant

Organic Electronic Materials

2020

Graduate course on design strategies for energy level tuning of organic materials. Responsible for preparing exams, and holding office hours.

Organic Chemistry Lab

2019

Undergraduate lab course in organic synthesis. Responsible for pre-lab experiment, preparing exams, delivering lectures, and tutoring organic synthesis and characterization.

General Chemistry Lab I&II

2017 - 2019

Undergraduate lab course in organic and inorganic chemical experiment.
Responsible for preparing experimental setups and exams, delivering lectures, and tutoring experimental details.